WORKSHOP · UV Amsterdam SER-AI

Measuring What We Compute

Towards (More) Efficient AI Use

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About me

Background

- PhD Candidate: NLP & Political Comm.
- Munich Data Science Institute
- TUM Climate Ventures
- MSc Big Data Analytics

Interests

- Data science, AI/NLP engineering
- AI & sustainability, data-intensive apps
- Creator of tidycarbon & fastcorpus
- Rock climber and daily dog walker

Agenda

Part 1: Eco Impacts of AI (50 min)

- Green AI as a social science topic
- Estimating emissions of AI
- Estimating emissions in social science
- Reduce, Reuse, Recycle

Part 2: Hands-on (40 min)

- Setup: Jupyter, Colab and uv
- Exercise 1: Using CodeCarbon + TidyCarbon
- Exercise 2 (maybe): Agent Skills

GITHUB REPO: https://github.com/seankellyhp/eco-measure-amsterdam

Python with uv, CodeCarbon, EcoLogits, Transformers, Jupyter. Optional R: TidyCarbon, Tidyverse, reticulate.

MIT Technology Review

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CLIMATE CHANGE AND ENERGY

Why Microsoft made a deal to help restart Three Mile Island

A once-shuttered nuclear plant could soon return to the grid.

By Casey Crownhart September 26, 2024



Amazon Tries to Boost Nuclear Power in Maryland After Scrapping Data Center Deal

The tech giant will help expand power generation at Constellation Energy's Calvert Cliffs plant and explore adding new reactors there

Amazon invests in nuclear energy project targeting nearly 1 gigawatt of carbon-free energy for Indiana

Amazon, X-energy, and AEP are collaborating to explore building small modular reactors in Indiana that would add carbon-free power to the regional grid.

Google enters massive 3.6-GW power deal with Constellation Energy

Sources: MIT Technology Review; Forbes; WSJ; Reuters (links in notes)

tom's HARDWARE

Elon Musk confirms xAI is buying an overseas power plant and shipping the whole thing to the U.S. to power its new data center — 1 million AI GPUs and up to 2 Gigawatts of power under one roof, equivalent to powering 1.9 million homes



Data center developers are building their own power plants

Data center developers have announced 90 GW of behind-the-meter projects, according to Cleanview's project tracker.

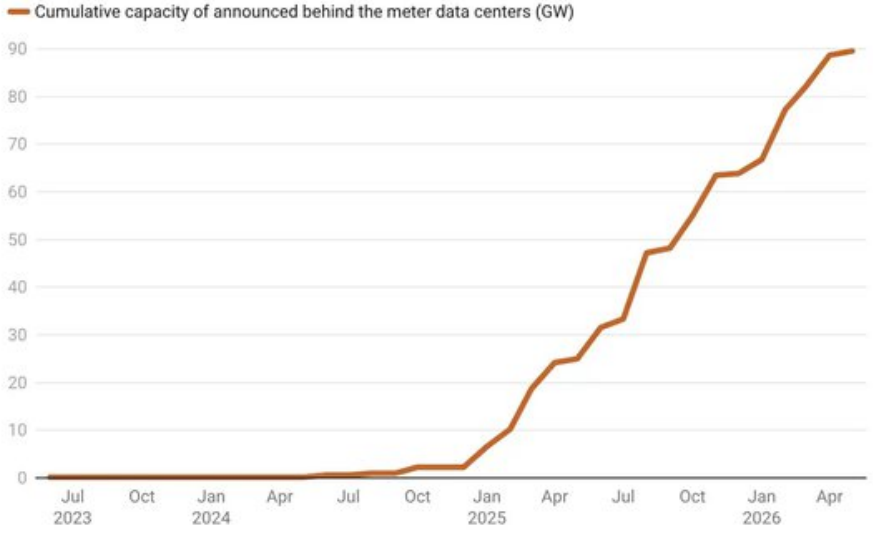


Chart: Michael Thomas • Source: Cleanview project tracker • Created with Datawrapper

Sources: Tom's Hardware; cleanview.co (behind-the-meter data centers)

Data centers in the United States

Cleanview tracks **4,592 data centers** on **1,532 campuses** in the US, many of them AI and hyperscale.

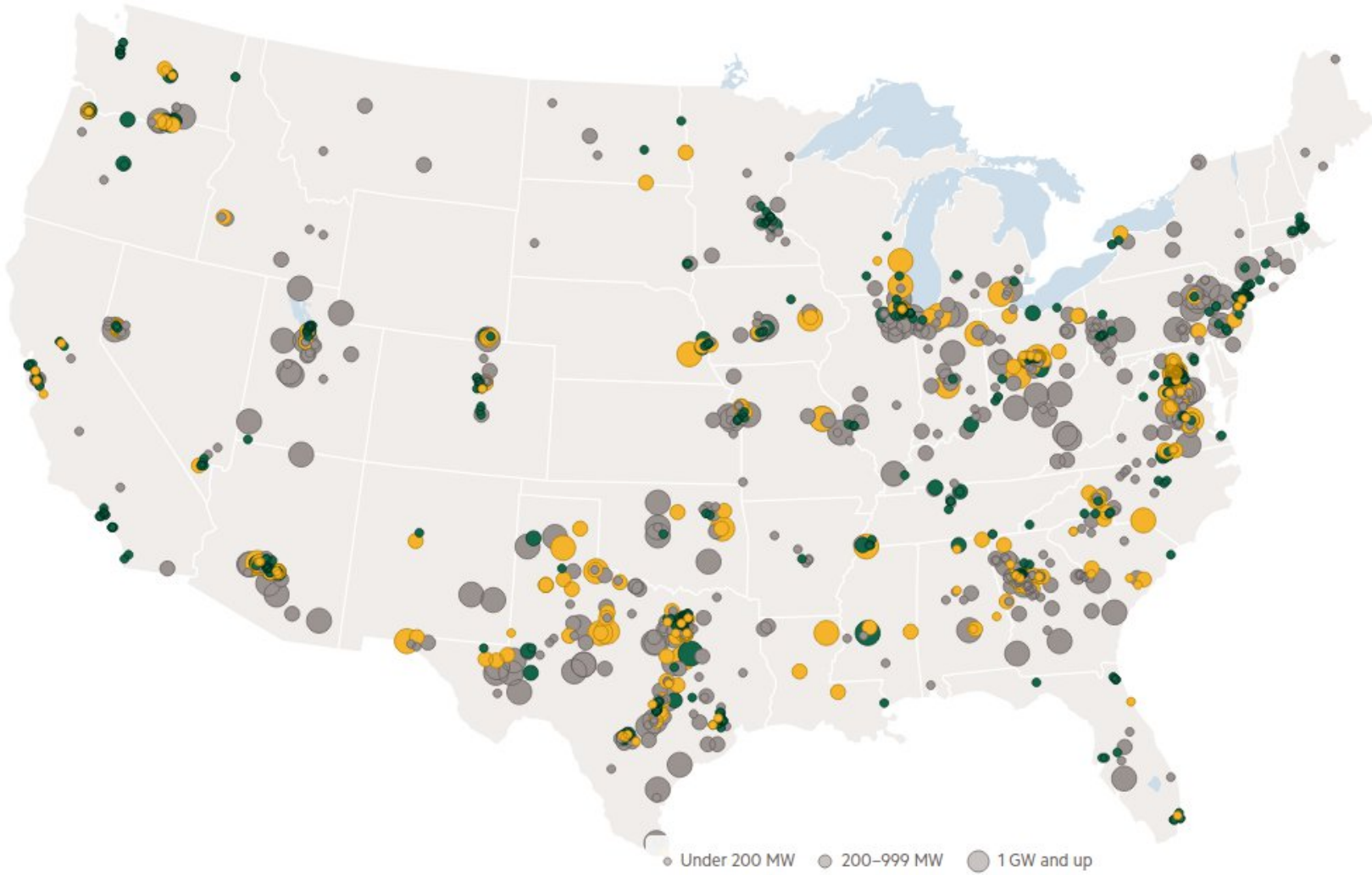
Campuses

Planned	**703**
Under construction	**274**
Operating	**555**

Data centers in your state

Choose a state

Updated daily. Last updated November 6, 2026.



Source: cleanview.co/data-centers/us









2026 data center ballot measure scorecard

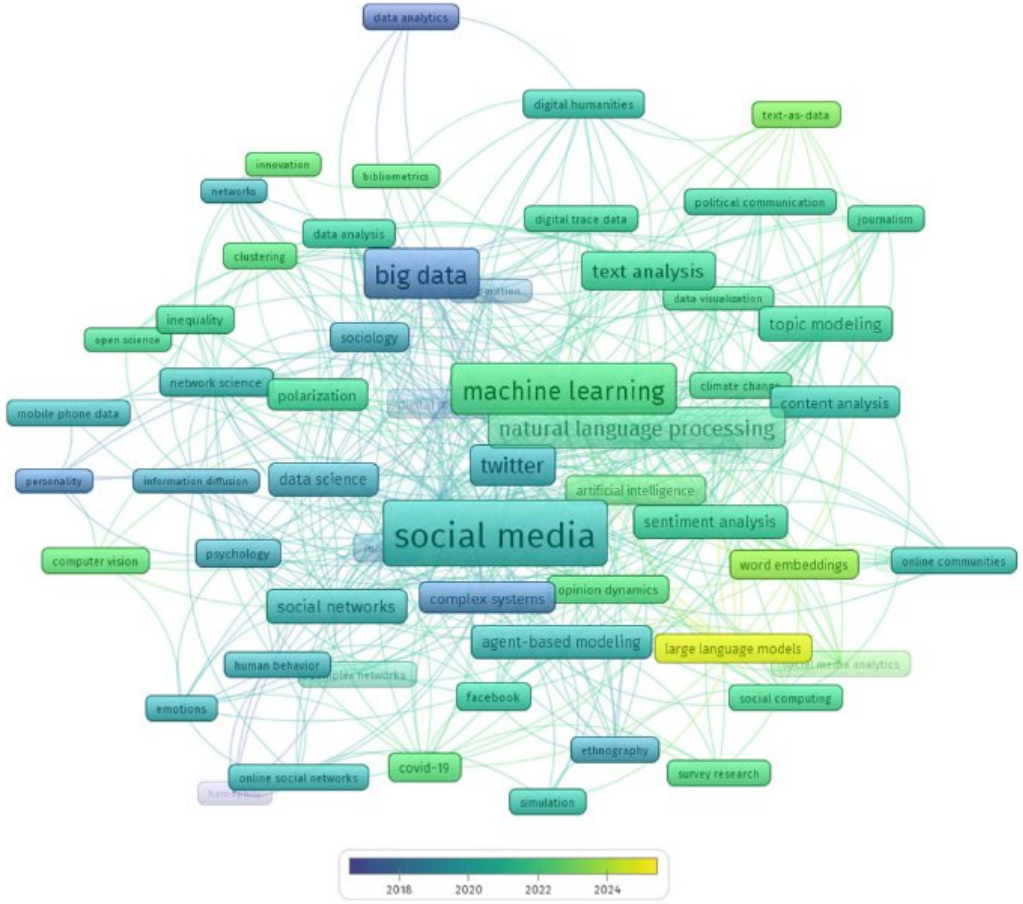
Outcomes favoring development		Outcomes limiting development
1	—	46
2% of decided measures		98% of decided measures

47 decided · three pending

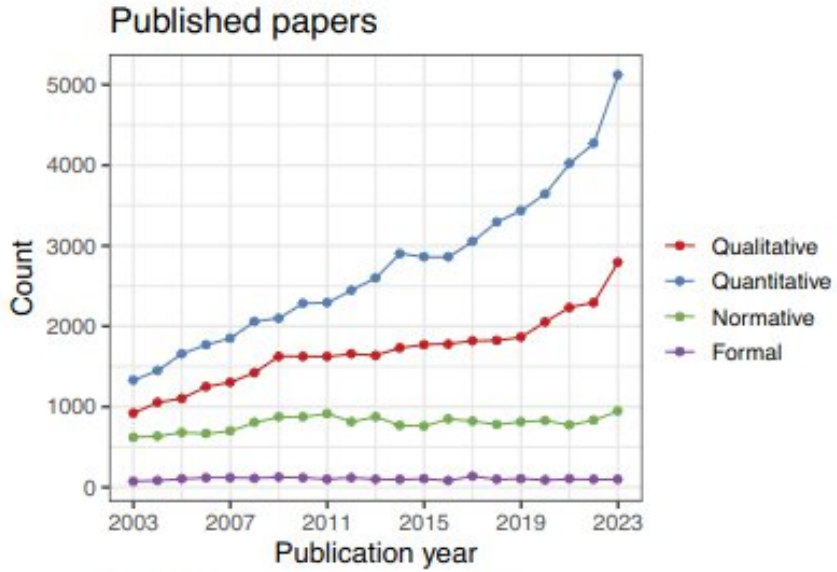
Source: MIT Technology Review, 4 Nov 2026

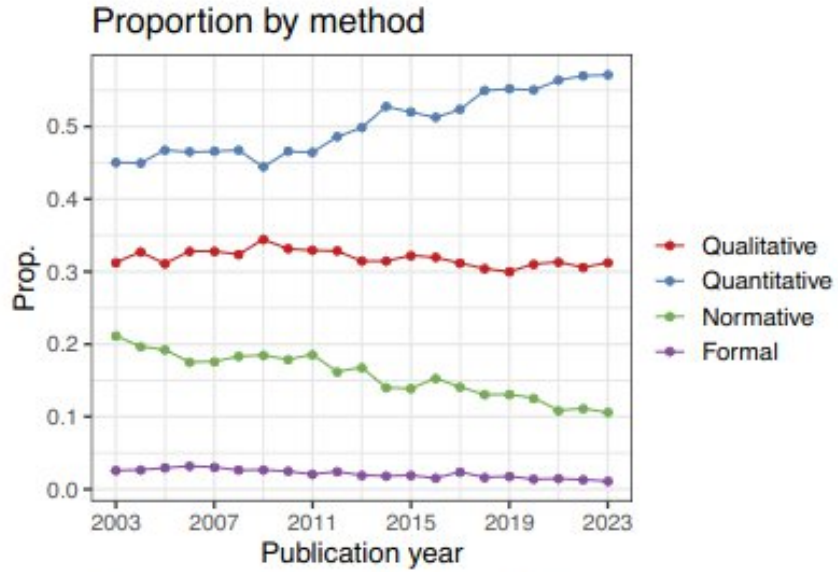
Green AI as a Social Science Topic

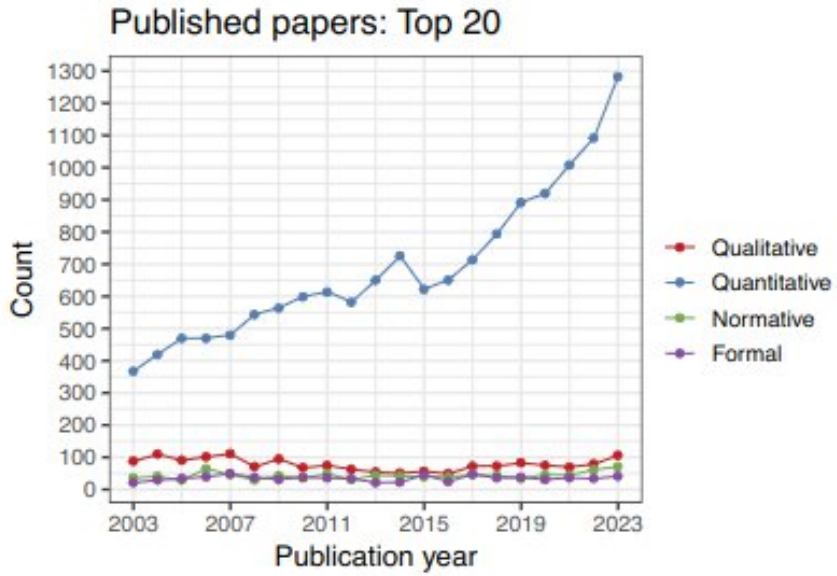
Computational Social Science

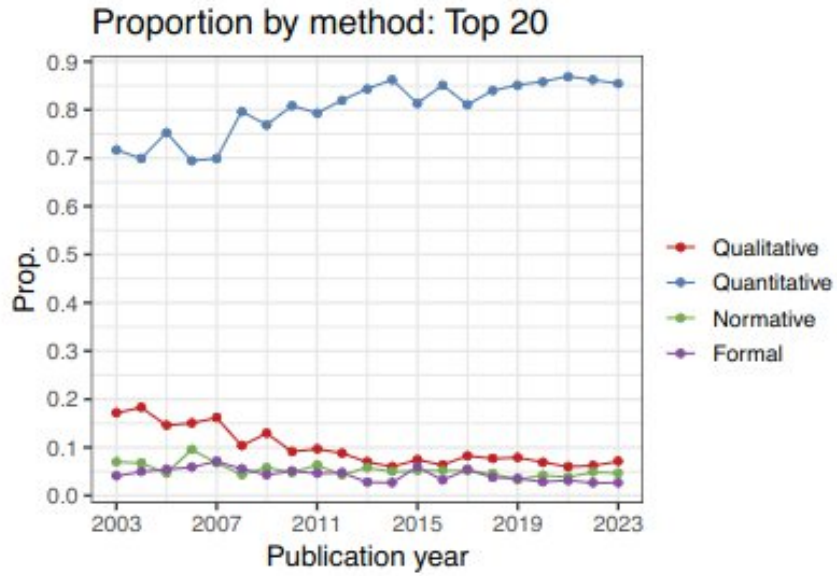


Are we just social scientists who use AI and machine learning?


Published papers
Count
Publication year
Qualitative
Quantitative
Normative
Formal


Proportion by method
Prop.
Publication year
Qualitative
Quantitative
Normative
Formal

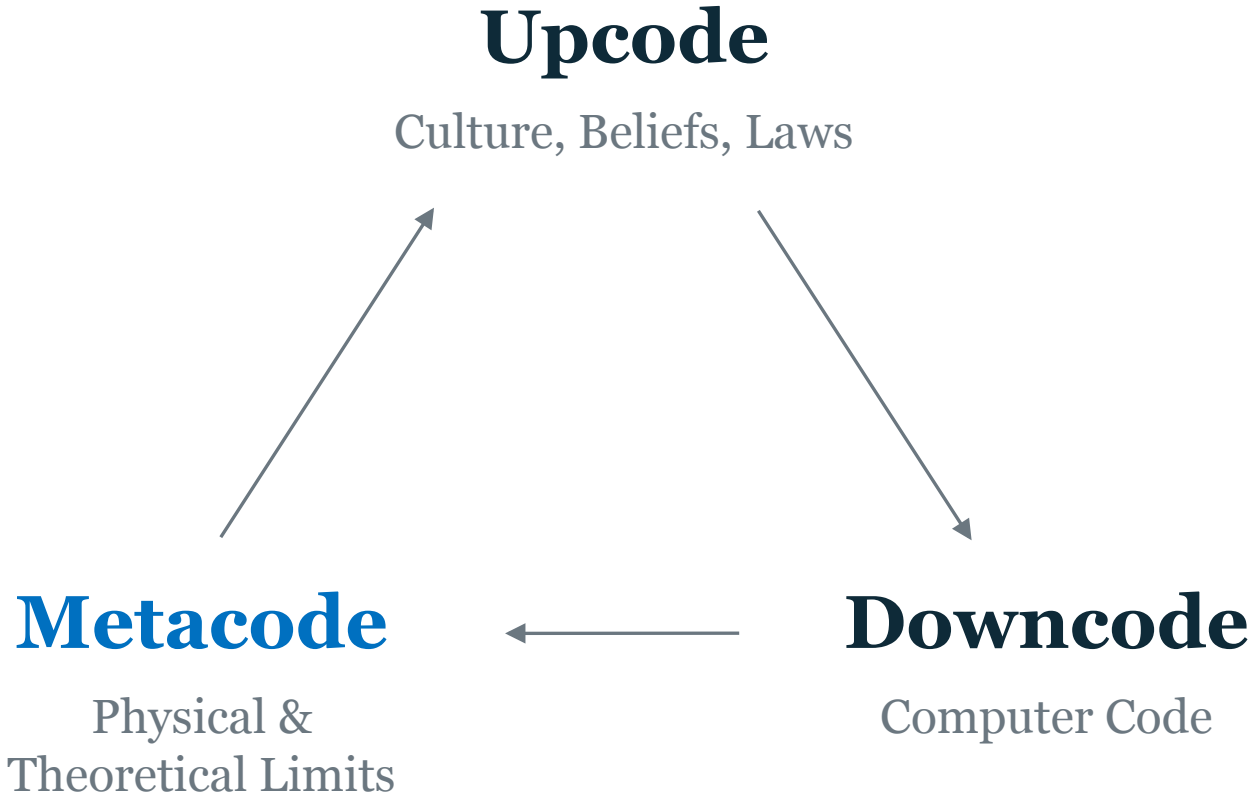

Published papers: Top 20
Count
Publication year
Qualitative
Quantitative
Normative
Formal

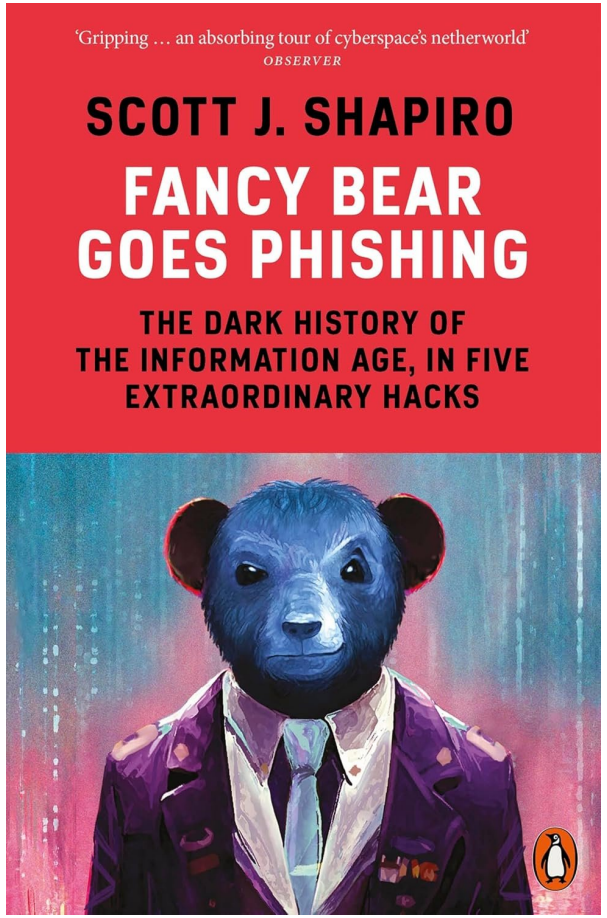

Proportion by method: Top 20
Prop.
Publication year
Qualitative
Quantitative
Normative
Formal

Critical CSS: AI fairness and social biases

- Automation may reproduce inequality, but perceived as more objective (Benjamin, 2019; Zuboff, 2019; O'Neil, 2016)
- Challenging technology as a neutral tool (Baden et al, 2020; Noble, 2018)
- Frame algorithmic bias as more than a technical issue (Hase et al., 2025; Kathirgamalingam et al., 2024)
- Adapt error-correction strategies, or leverage bias as a measurement tool (Waldherr et al, 2025; van Atteveldt & Peng, 2018)

CSS, Complexity and Sustainability





Shapiro 2023; Waldherr et al., 2021; Turing, 1937

What is Green AI?

Red AI

Buy novel results using computational power.

+ **Model Complexity**
+ **Data Examples**
+ **Hyperparameter Experiments**

Green AI

Novel results at lower equal computational cost.

— **Carbon Emissions**
— **Electricity Usage**
— **Compute Time**
— **Model Parameters**
— **Floating Point Operations**

Schwartz et al., 2019

Estimating Emissions of AI



Tench Island, Papa New Guinea



Saying ‘Please’ and ‘Thank You’ to ChatGPT Costs OpenAI ‘Tens of Millions of Dollars’

It’s more environmentally friendly (and cost-efficient) to skip the niceties.

Pole to Pole with Will Smith, National Geographic, 2026

What is a kWh?

Power is a rate (watts). Energy is power × time: **1,000 W for one hour = 1 kWh.**

Unit	Example	Estimate
Wh	2 minutes 10-watt LED Bulb	**0.3 Wh**
Wh	One average LLM Chat query-response (Gemini – GPT 5.2)	**0.24 - 0.6 Wh**
Wh	One hour Claude Code session (Opus 4.8)	**53.4 Wh**
kWh (1 thousand)	Hourly US household electricity per household	**1.223 kWh**
kWh	Annual average US household	**10,791 kWh**
MWh (1 million)	Training Aleph-Alpha Kolibri-1 (78 B Params)	**950 MWh**
GWh (1 billion)	Training GPT-4	**50–70 GWh**
TWh (1 trillion)	All data centers in 2025	**448 TWh**

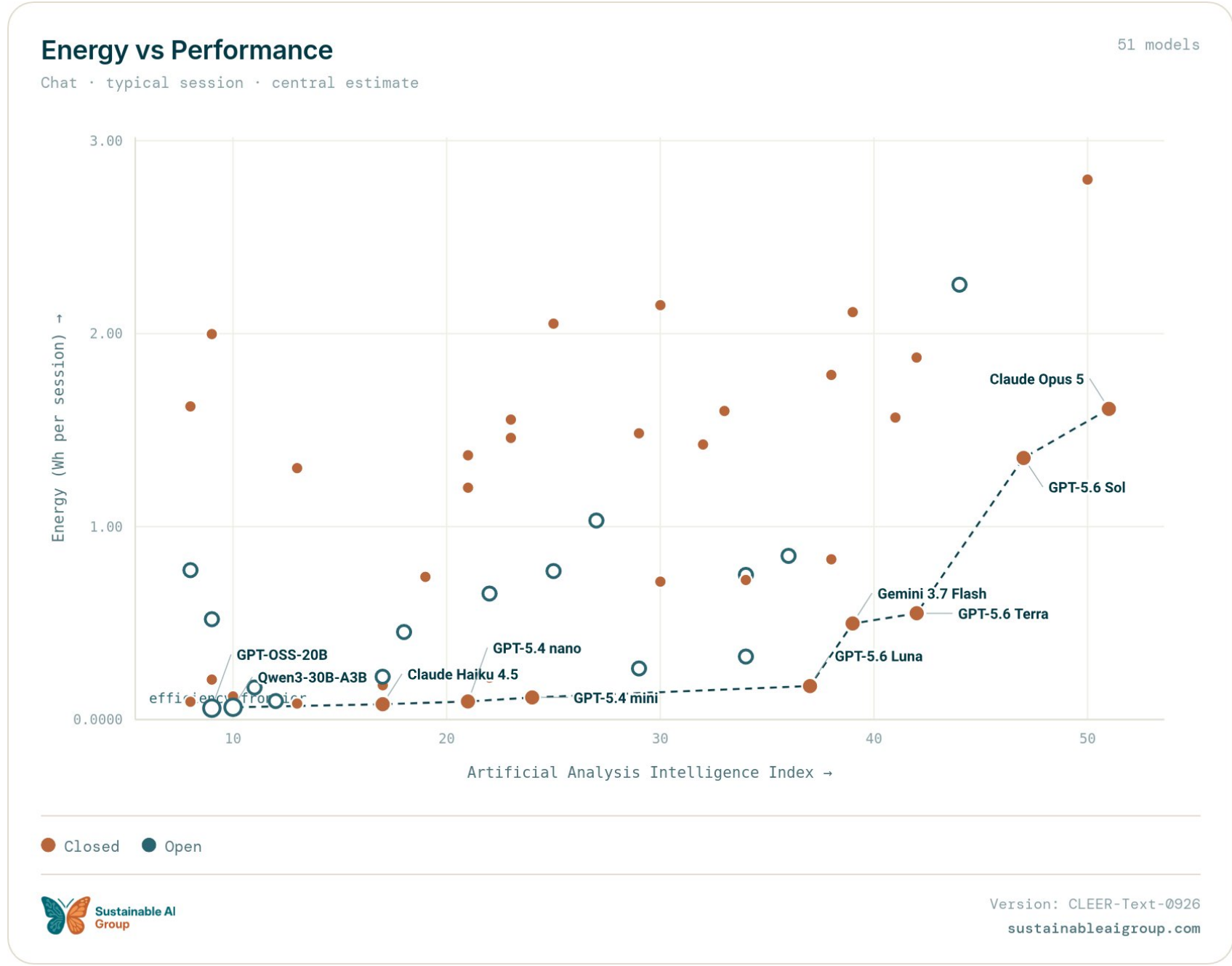
The AI openness debate

Open weights

- Self host, direct measure
- Released code and models spare the emissions of re-training
- Non-optimized serving environments

Proprietary models

- No direct access, estimates only
- Often lack per-request energy or emissions data
- Model updates can change behavior without notice



Source: cleerdash.sustainableaigroup.com

How energy and emissions are calculated

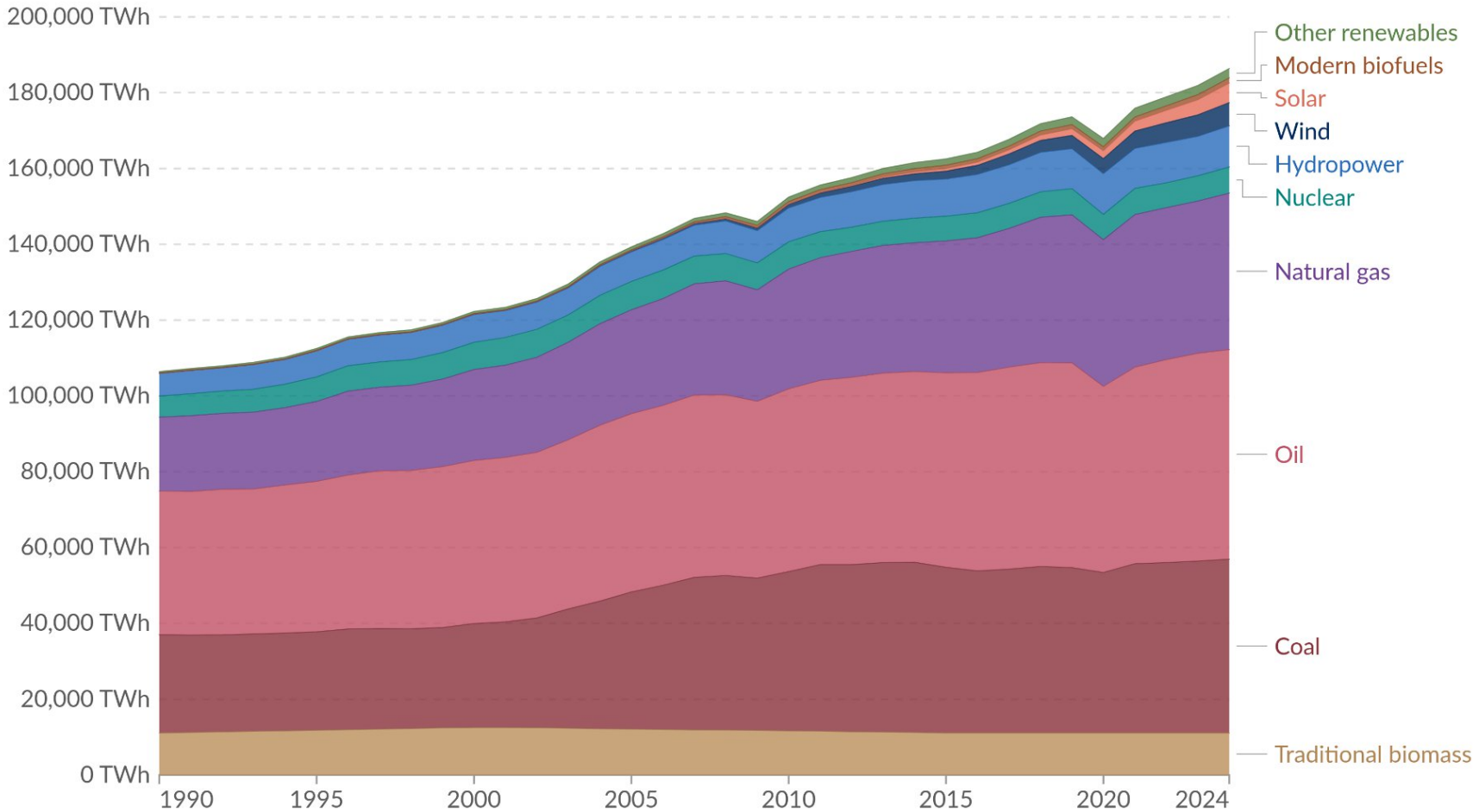
$$\sum \left(\text{Power} \times \text{Time} \times \text{PUE} \right) \times \text{Grid intensity} = \textbf{Emissions}$$

Power	Time	PUE	Grid intensity	Emissions
GPU + CPU + RAM, read while the code runs	How long the code runs in real time, sample intervals	Power Usage Effectiveness. Data-center overhead	gCO_2e per kWh where the power is drawn	gCO_2e for the task

https://docs.codecarbon.io/latest/explanation/methodology/

Global primary energy consumption by source

Primary energy[1] is based on the substitution method[2] and measured in terawatt-hours[3].



Data source: Energy Institute - Statistical Review of World Energy (2025); Smil (2017)

OurWorldinData.org/energy | CC BY

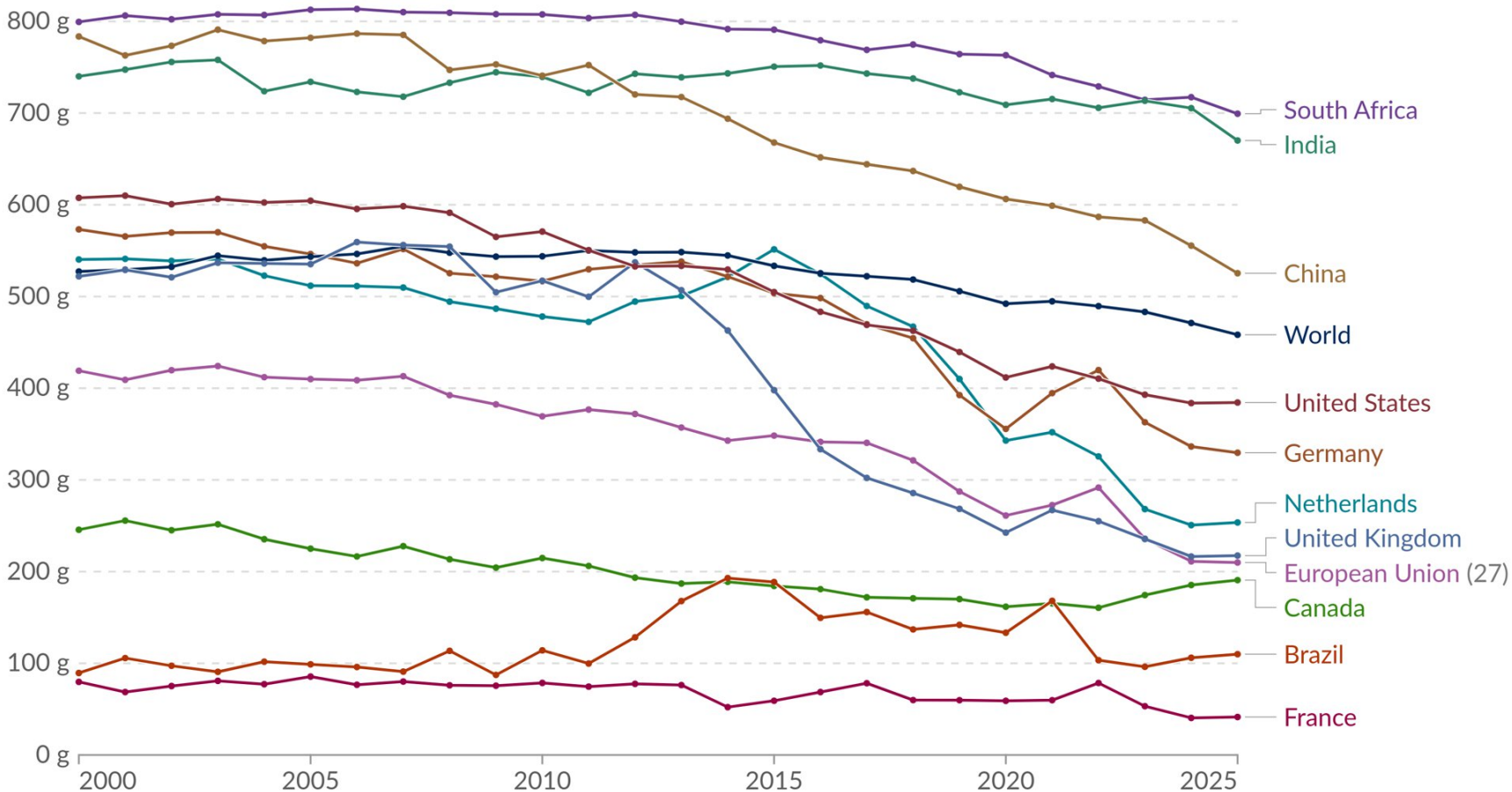
Note: In the absence of more recent data, traditional biomass is assumed constant since 2015.

Source: Our World in Data

Lifecycle carbon intensity of electricity 2000 to 2025

Our World in Data

Measured in grams of carbon dioxide-equivalents[1] emitted per kilowatt-hour[2] of electricity generated. Emissions are estimated on a lifecycle basis, including upstream, supply chain and manufacturing stages, and cover all greenhouse gases.



Data source: Ember (2026)

OurWorldinData.org/electricity-mix | CC BY

Source: Our World in Data

Estimating Emissions in Social Science

Research questions

RQ1 How much energy is consumed when using LLMs for text analysis tasks compared to using lightweight alternative methods?

RQ2 What is the optimum balance between LLM size and task performance when completing text analysis tasks?

Methodology: Comparing LLMs to Non-LLM Methods

Tasks

Text analysis:

- Sentiment Analysis
- Multi-Class Classification
- Named-Entity-Recognition

COMMUNICATION METHODS AND MEASURES
https://doi.org/10.1080/19312458.2024.2383453

Routledge
Taylor & Francis Group

OPEN ACCESS Check for updates

JST and rJST: joint estimation of sentiment and topics in textual data using a semi-supervised approach

Christian Pipal[a], Martijn Schoonvelde[b], Gijs Schumacher[c], and Max Boiten[d]

Do AIs know what the most important issue is? Using language models to code open-text social survey responses at scale

Research and Politics
January-March 2024: 1–7
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DOI: 10.1177/20531680241231468
journals.sagepub.com/home/rap
Sage

Jonathan Mellon[1], Jack Bailey[2], Ralph Scott[3], James Breckwoldt[2], Marta Miori[2] and Phillip Schmedeman[1]

COMMUNICATION METHODS AND MEASURES
2024, VOL. 18, NO. 4, 371–389
https://doi.org/10.1080/19312458.2024.2324789

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Automatically Finding Actors in Texts: A Performance Review of Multilingual Named Entity Recognition Tools

Paul Balluff, Hajo G. Boomgaarden, and Annie Waldherr

Hamilton-Palicki, Bravo & Claessen, 2026

Methodology: Comparing LLMs to Non-LLM Methods

Tasks

Text analysis:
- Sentiment Analysis
- Multi-Class Classification
- Named-Entity-Recognition

Models

LLMs (Quantized, Local):
- Mistral Nemo - 12B
- Gemma 3 - 24B
- Mistral Small - 27B
- Deepseek R1 - 32B

Non-LLMs:
- Dictionaries
- SVM
- NER Libraries
- BERT (Pretrained/Fine-Tuned)

Hamilton-Palicki, Bravo & Claessen, 2026

Methodology: Comparing LLMs to Non-LLM Methods

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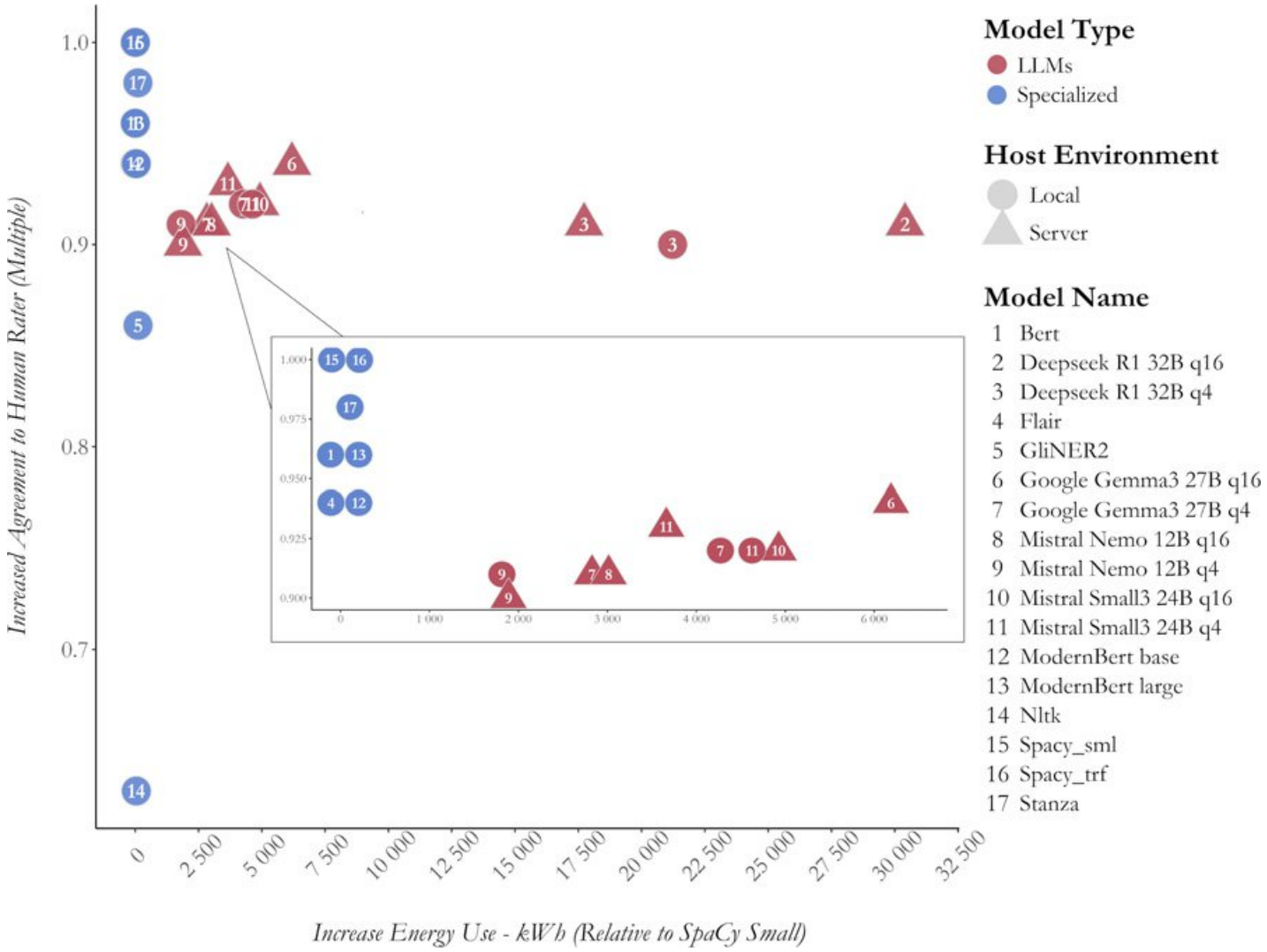
Metrics

Human-Rater Agreement:
- Pearson Correlation, Percent Agreement

Model Effectiveness:
- F1, Precision, Recall, Duration

Sustainability:
- kWh Energy Use, CO_2 Emissions

Bigger, not always better: Named Entity Recognition



- Task-specific models (blue): high agreement, almost no energy
- LLMs (red) use thousands of times more energy
- Deepseek-R1 uses up to 30,000× the energy of SpaCy Small, with lower agreement
- Sentiment and Classification fare better for LLMs

Hamilton-Palicki, Bravo & Claessen, 2026

Local vs. server deployment

Sentiment, 200 speeches (4-bit)	Local: consumer GPU, < 24 GB	Server: < 80 GB
Gemma 3 27B	0.12 kWh · 14.0 min	0.10 kWh · 11.7 min
Mistral Small 3 24B	0.12 kWh · 13.5 min	0.07 kWh · 7.3 min
DeepSeek-R1 32B	0.34 kWh · 39.4 min	0.28 kWh · 27.7 min
Mistral Nemo 12B	**0.02 kWh · 3.3 min**	0.03 kWh · 4.0 min

- Server-grade hardware is more energy efficient than consumer-grade hardware
- Model choice matters more than hardware efficiency
- Self-hosting gives the tracker full access to the hardware

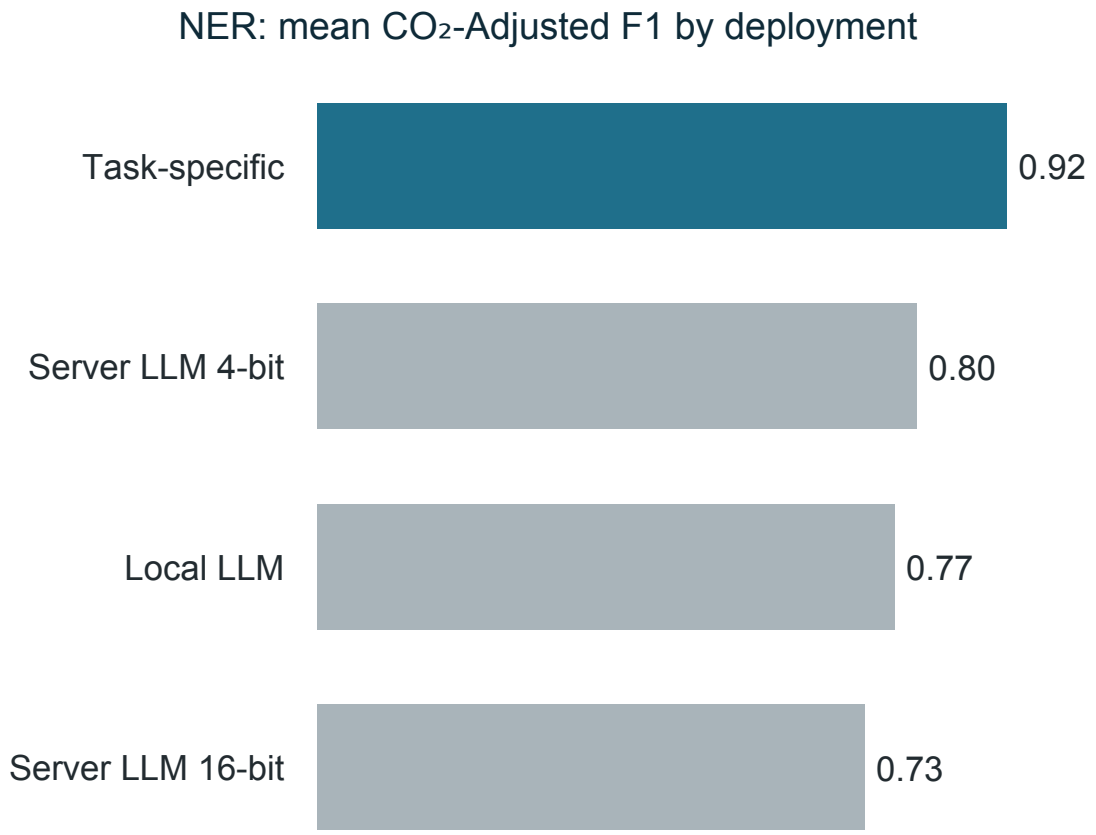
Hamilton-Palicki, Bravo & Claessen, 2026

CO_2-Adjusted F1 Score

$$F1_{adj} = F1 \times \exp(-K \cdot E / N)$$

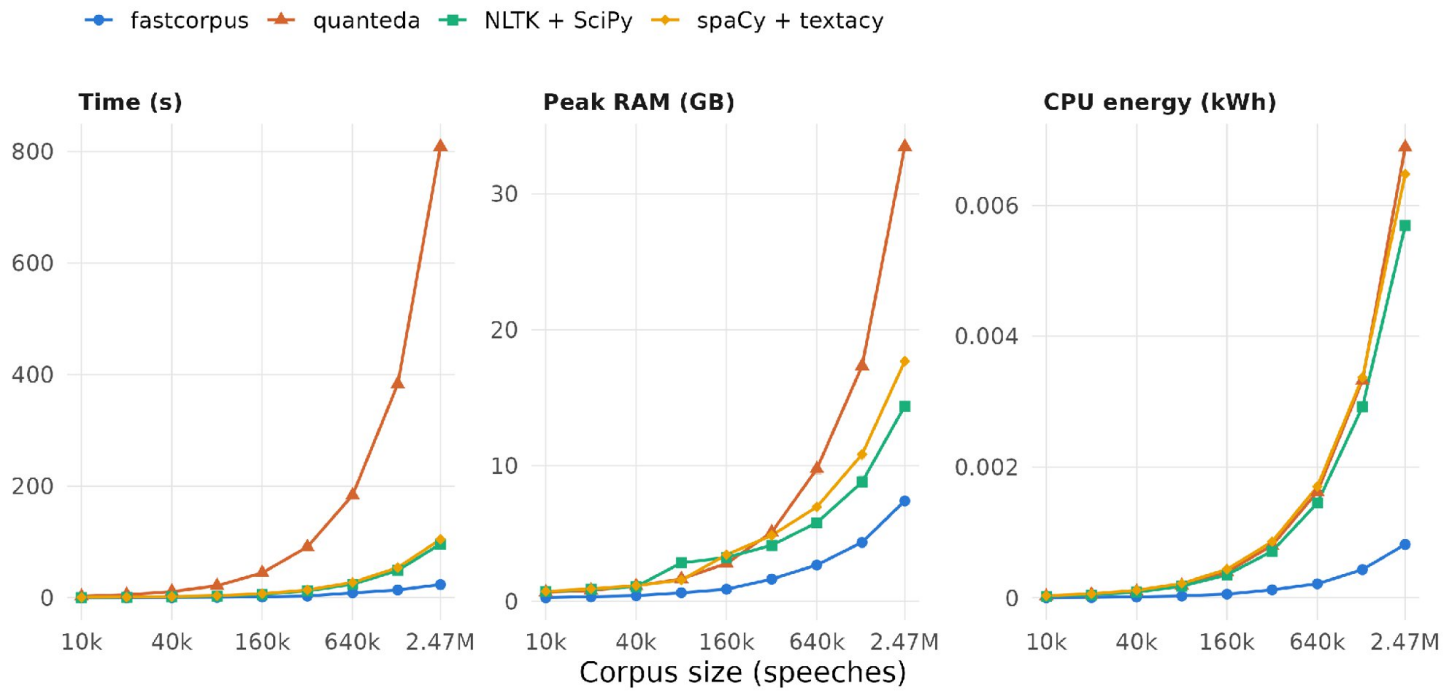
E = grams CO_2e for the task, N = sample size, K = penalty (1 = balanced)

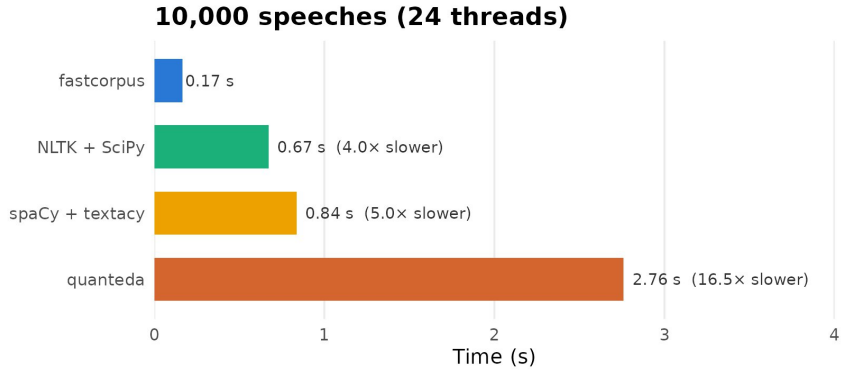
- Exponential "Safe zone" for low-emission models, then fast decay for power hungry models
- Small task-specific models (spaCy, fine-tuned BERT) may reach a Pareto-optimal balance of performance and energy

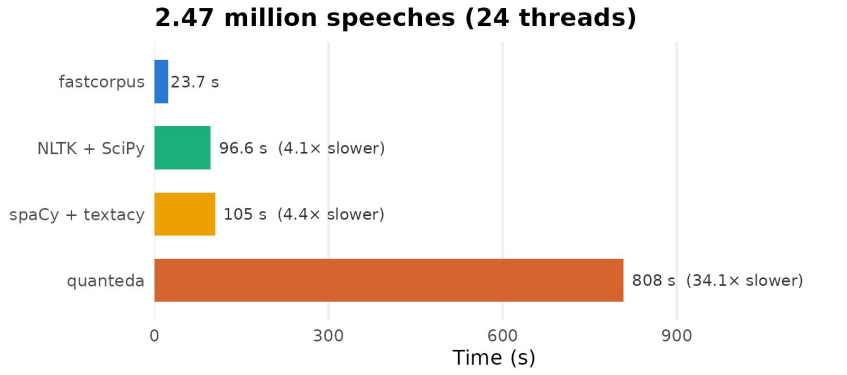


Hamilton-Palicki, Bravo & Claessen, 2026

FastCorpus: Fast, Efficient & Sustainable NLP







Reduce, Reuse, Recycle

Data Centers and Jevons Paradox

205 TWh

2018, about 1% of global electricity
(Masanet et al. 2020)

448 TWh

2025; ranked as a country, 11th in the world
(UNU 2026)

945 TWh

2030 projection, almost 3%; AI ≈ 40%
(UNU 2026)

Increased efficiency

- Research over-estimates resource consumption of closed source AI models.
- Energy per median Gemini prompt fell 33× in twelve months (Elsworth et al., 2025)

Increased use

- Jevons Paradox: cheaper coal use expanded coal use
- Rebound effects are direct and indirect (Luccioni et al., 2025)
- Net zero by 2030 is unlikely for US AI servers (Xiao et al., 2025)

Lifecycle of AI

Extraction	Manufacturing	Training	Inference	End of life
Minerals for chips, servers, batteries	Most hardware carbon; ultrapure water	Plus experiments and tuning	Repeated at scale	E-waste, rarely counted

- Doubles AI emission estimates compared with training alone (Luccioni et al., 2024)
- Meta Llama 4: 30% data, 30% training, 40% inference (Chien et al., 2026)
- Unknown carbon costs GPU manufacturing (Luccioni et al., 2024)
- Hazardous mining and e-waste impacts communities in the Global South (UNU 2026, Crawford, 2021)

Minimizing Footprint

01
AI Energy Score

Standardized benchmarks for models on fixed tasks and hardware

02
Special vs General

Know your task and pre-trained models; use tools.

03
Smaller Models

Distillation, quantization, local serving

04
Where and When?

The same job emits up to 30× less in a cleaner grid or better hardware

05
CodeCarbon*

Python tracker for CPU, GPU and RAM energy, CO_2e and Water

06
EcoLogits*

Estimates energy, warming and resource use per API request to closed models

*CodeCarbon and EcoLogits are dependencies of TidyCarbon

Part 2: Hands-On

Measuring What you Compute

- Set up Python jupyter notebooks (CodeCarbon, Ecologits)
  - Getting Started
  - Sentiment Tracking
  - LLM Tracking
- Jupyter Lab locally (using uv) or Google Colab (recommended)
- TidyCarbon notebook available for R

GITHUB REPO: https://github.com/seankellyhp/eco-measure-amsterdam

How energy and emissions are calculated

$$\sum \left(\text{Power} \times \text{Time} \times \text{PUE} \right) \times \text{Grid intensity} = \textbf{Emissions}$$

Power	Time	PUE	Grid intensity	Emissions
GPU + CPU + RAM, read while the code runs	How long the code runs in real time, sample intervals	Power Usage Effectiveness. Data-center overhead	gCO_2e per kWh where the power is drawn	gCO_2e for the task

https://docs.codecarbon.io/latest/explanation/methodology/

CodeCarbon (Py): Measurements of Hardware

EmissionsTracker or EmissionsTrackerOffline

```
from codecarbon import EmissionsTracker

with EmissionsTracker(project_name="my-first-tracking") as tracker:
    # Your code here
    total = 0
    for i in range(10_000_000):
        total += i

print(f"Computation result: {total}")
```

Track complete programs with tracker.start then tracker.stop
Track code blocks with tracker.start_task then tracker.stop_task

EcoLogits (Py): Estimates of AI Providers

Estimate impacts for selected providers and models (Anthropic, OpenAI, Mistral, Google Gemini and more)

```
from ecologits import EcoLogits
from openai import OpenAI

EcoLogits.init(providers=["openai"])

client = OpenAI(api_key="<OPENAI_API_KEY>")

response = client.chat.completions.create(

        model="gpt-4o-mini",

        messages=[{"role": "user", "content": "Tell me a funny joke!"}])
```

EcoLogits (Py): Choices of Estimates

Get estimates with response.energy.value or with llm_impacts()

energy = Total energy consumed (electricity)

gwp = Global warming potential; Total GHG emissions

adpe = Total abiotic resource depletion

pe = Total energy consumed from primary sources

wcf = Water consumption footprint

TidyCarbon: A Tidy R Wrapper

Track the energy use of an R program

```
tracker <-
tidycarbon::carbon_init(
  project_name = "GREEN AI",
  measure_power_secs = 1,
  output_file = "emissions.csv")

tidycarbon::tracker_start(tracker
)

#
# YOUR BEST R CODE HERE
#

tidycarbon::tracker_stop(tracker)
```

Integrate into an analysis pipeline

```
tokens(big_corpus,
  remove_punct = TRUE,
  remove_numbers = TRUE) |>
  tokens_tolower() |>
  tokens_remove(stopwords("en"))
|>
  tidycarbon::carbon_step(
    tokens_ngrams(n = 1:3)) |>
  tidycarbon::carbon_step(
    dfm())
```

Benchmark alternative approaches

```
tidycarbon::carbon_bench(
  LLM = make_query(
    text = txt,
    prompt = "Is this text:
      positive, neutral,
      or negative?"),
  DICT = textstat_polarity(
    text = txt,
    dictionary = LSD2015),
  times = 10) |>
  autoplot(metric =
"emissions_total")
```

`tidycarbon::carbon_init_remote` for Ecologits

Extra Slides

Water Consumption

0.18–1.1 L

of cooling water per kWh, depending on the data center (Luccioni et al., 2024)

500 mL

per 10–50 GPT-3 responses
(Li et al., 2023)

4.5 trillion L

used by data centers in 2025
(UNU 2026)

Mineral Use

100s of tons

of ore dug up and processed to get one ton of a common metal such as copper
(Luccioni et al., 2024)

Conflict

Cobalt and tungsten are mined or traded in conflict areas, sustaining human rights abuses (Crawford, 2021)

Weak rules

Mining is energy-, water- and land-intensive, often where protections are weaker (UNU 2026)

Afterlife: E-Waste

2.5 Mt

e-waste a year from AI infra by 2030, 250 Eiffel Towers (UNU 2026)

3–4 years

typical life of a data-center server CPU
(Gupta et al., 2020)

10–20%

data-center emissions from recycled or older components (Chien et al., 2026)